

[14] Analysis of Total Phenols and Other Oxidation Substrates and Antioxidants by Means of Folin–Ciocalteu Reagent

By VERNON L. SINGLETON, RUDOLF ORTHOFER,
and ROSA M. LAMUELA-RAVENTÓS

Introduction

Phenols occurring in nature and the environment are of interest from many viewpoints (antioxidants, astringency, bitterness, browning reactions, color, oxidation substrates, protein constituents, etc.). In addition to simple benzene derivatives, the group includes hydroxycinnamates, tocopherols, and flavonoids in plants and foods from them, tyrosine and DOPA derivatives in animal products, and additives such as propyl gallate in foods. Estimation of these compounds as a group can be very informative, but obviously not simple to accomplish. Isolative methods such as high-performance liquid chromatography (HPLC) are difficult to apply to such a diverse group having, furthermore, many individual compounds within each subgroup. Interpretation of such results is even more difficult.

Phenols are responsible for the majority of the oxygen capacity in most plant-derived products, such as wine. With a few exceptions such as carotene, the antioxidants in foods are phenols. Among those added to prevent oxidative rancidity in fats are the monophenols (benzene derivatives with a single free hydroxyl group) 2,6-di-*tert*-butyl-4-hydroxytoluene (BHT) and its monobutylated anisole analog (BHA). *tert*-Butyl substituents function mainly to increase the lipid solubility. In aqueous solution the parent monophenols and others can also function as antioxidants. Therefore, it is important that total phenol assays include monophenols as well as more easily oxidized polyphenols.

An antioxidant effect can be from competitive consumption of the oxidant, thus sparing the target molecules being protected, and from quenching the chain reaction propagating free radical oxidation. Antioxidants become oxidized as they interfere with the oxidation of lipids and other species. Paradoxically, because of coproduction of hydrogen peroxide as an antioxidant phenol or ascorbic acid reacts with oxygen, coupled oxidation can occur of substrates (ethanol, for example) that would not react readily with oxygen alone.¹

¹ H. L. Wildenradt and V. L. Singleton, *Am. J. Enol. Vitic.* **25**, 119 (1974).

If one electron is removed (oxidized) from a phenolate anion, the product is a semiquinone free radical. Removal of a second electron from *ortho*- or *para*-diphenols produces a quinone. A mixture of phenol and quinone equilibrates to produce semiquinone intermediates. The molecule accepting a removed electron is, of course, reduced. Free radicals are very reactive molecules with an unpaired electron. Encountering another free radical from any source (its own type, lipoidal, etc.), the two combine to form a new covalent bond, terminating any chain reaction caused by extraction by the free radical of an electron from an intact molecule to generate another free radical. The unpaired electron in a semiquinone can resonate among the former hydroxyl and the positions *ortho* and *para* to it (two, four, or six of the ring). A mixture of dimerized products results as the new bonds form. If the new bond is to one of the ring carbons, the phenolate is regenerated. Oxidation may then not only be repeated, but the regenerated phenol is often oxidized more easily than the original one. If the important property of oxidizability is to be the basis for the quantitation of phenols, the reaction must be brought quickly to a conclusion to minimize such regenerative polymerization.

That the phenolate ion is important is shown by the fact that the uptake of oxygen by phenols can be rapidly complete near or above the pK of the phenol (usually about pH 10).²⁻⁴ Because of the relative ease of removing an electron from its phenolate, the less acidic the particular phenol the easier its oxidation. At lower pH the reaction appears proportionate to the pH, but as low as pH 3 equilibria supply enough phenolate among natural phenols of low acidity to allow slow reaction (with, of course, additional total oxygen uptake from regenerative polymerization and any other slow, competing reaction). Reaction at alkaline pH is indicated for assay purposes.

A method based on these considerations can be very useful provided it is reproducible, its basis understood, and its applicability verified. The proper use of the reagent proposed by Otto Folin and Vintila Ciocalteu⁵ is such a method. The resultant total value is often directly comparable and informative among different samples, e.g., the total phenol content of commercial wines can range from about 50 to 5000 mg/liter.⁶ This 100-fold range not only distinguishes white, pink, and red wines as groups, but enables the evaluation of high versus low astringency, browning tendency,

² J. A. Rossi, Jr. and V. L. Singleton, *Am. J. Enol. Vitic.* **17**, 231 (1966).

³ V. L. Singleton, *Am. J. Enol. Vitic.* **38**, 69 (1987).

⁴ J. J. L. Cillers and V. L. Singleton, *J. Agric. Food Chem.* **37**, 890 (1989).

⁵ O. Folin and V. Ciocalteu, *J. Biol. Chem.* **73**, 627 (1927).

⁶ V. L. Singleton and P. Esau, *Adv. Food Res., Suppl.* **1**, 1 (1969).

and other characteristics within a group. Determination of the total content before and after a treatment to remove or inactivate specific subgroups of reactants can give more specific information. Determination of single substances by HPLC and calculation of the contribution of that content to the total can lead to a balance sheet of contributors to the total and indicate the magnitude of any remaining unknown fraction.² This latter adaptation appears capable of more exploitation than has been made so far.

Analyses made with reagents of the Folin and Ciocalteu (FC) type are often numerically appreciably different than those obtained with other methods purported to determine total phenols. Nevertheless, relative values usually correlate well among these methods, as long as samples of similar type are being compared. This correlation may be somewhat illusory and not found among samples of widely different types because qualitatively and relatively the particular mixture of different positive reactants may be rather constant in samples of a given product.

Considering the heterogeneity of natural phenols and the possibility of interference from other readily oxidized substances, it is not surprising that several methods have been used for total phenol determination and none are perfect. Among such methods competing with FC are permanganate titration, colorimetry with iron salts, and ultraviolet absorbance. Oxidation with potassium permanganate is more difficult to standardize among different analysts and is subject to greater interferences, particularly from sugars. Several direct comparisons of FC methods with those based on KMnO_4 have shown the preferability of the FC. Colorimetry with iron salts has the problem, from the viewpoint of total phenol determination, that monophenols generally do not react and under some conditions vicinal diphenols and vicinal triphenols give different colors. Because of apparently less interference from dextrans, melanoidins, and proteins, ferrous colorimetry is often used for beer analysis, particularly heavy, dark beers. In most other direct comparisons FC has been found preferable.

Ultraviolet absorbance is difficult to apply to total phenol analysis not only because of potential interference from other compounds, which absorb at similar maxima, but because individual natural phenols differ greatly in both wavelength of maximum absorbance and their molar absorbance.⁷

Reagent

Folin and Denis⁸ first proposed a heteropoly phosphotungstate-molybdate reagent to react with tyrosine to give a blue color proportionate

⁷ A. Scalbert, in "Plant Polyphenols" (R. W. Hemingway and P. E. Laks, eds.), p. 259. Plenum Press, New York, 1992.

⁸ O. Folin and W. Denis, *J. Biol. Chem.* **12**, 239 (1912).

to the protein content. Application of this reagent is troubled by the occasional production of a white precipitate that interferes with direct colorimetry. It can be removed, but an extra step is required and some blue may be adsorbed on a filter. Improvements by Folin and Ciocalteu increased the proportion of molybdate and prevented this precipitation by adding lithium sulfate to the reagent. Comparison of the Folin-Denis (FD) procedure with that of Folin-Ciocalteu gives somewhat greater sensitivity and reproducibility for the FC.⁹ Nevertheless, in consideration of the desirability of constant procedures for official regulatory purposes, the FD reagent is sometimes still used.¹⁰ With proper technique and standards, results are very similar as the chemical basis is the same.

To prepare the FC reagent (FCR),^{5,9} dissolve 100 g of sodium tungstate ($\text{Na}_2\text{WO}_4 \cdot 2\text{H}_2\text{O}$) and 25 g sodium molybdate ($\text{Na}_2\text{MoO}_4 \cdot 2\text{H}_2\text{O}$) in about 700 ml of distilled water. Add 100 ml of concentrated HCl and 50 ml of 85% phosphoric acid. Boil under reflux for 10 hr (this time should not be shortened appreciably, but need not be continuous). Stop the heating and rinse down the condenser with a small amount of water and dissolve 150 g of $\text{Li}_2\text{SO}_4 \cdot 4\text{H}_2\text{O}$ in the slightly cooled solution. The resultant solution should be clear and intense yellow without a trace of green (blue). Any blue results from traces of reduced reagent and will cause elevated blanks. Refluxing for a short time after adding a couple of drops of bromine followed by removal of the excess bromine by open boiling (in a hood, of course) will correct this problem. If excess is avoided, a small amount of 30% hydrogen peroxide can be substituted for the bromine. Make the final solution to 1 liter. Filtration through a sintered glass filter removes insolubles, if necessary. Commercially prepared FCR is often employed. If protected from reductants, the reagent is ordinarily stable indefinitely, even if diluted.

Procedure

The manual method⁹ calls for 1.00 ml of sample, blank, or standard in water (or dilute aqueous solution) added to at least 60 ml of distilled water in a 100-ml volumetric flask. Add FCR (5.0 ml) and mix. After 1 min and before 8 min, add 15 ml of 20% sodium carbonate solution, adjust the volume to 100.0 ml, and read the color generated after about 2 hr at about 23° at 760 nm in a 1-cm cuvette. Provided appropriate standards and blanks are employed, considerable variation in these conditions may be permissi-

⁹ V. L. Singleton and J. A. Rossi, Jr., *Am. J. Enol. Vitic.* **16**, 144 (1965).

¹⁰ P. Cunniff, ed., "Official Methods of Analysis of AOAC International," 16th ed. AOAC International, Gaithersburg, MD, 1995.

ble. Important considerations are adequate FCR to react completely and rapidly with the oxidizable substances in the samples, sufficient time and mixing of the sample and the FCR solution before adding the alkali solution, and similar time/temperature conditions of color development.

Originally, saturated sodium carbonate was used for the alkaline reagent, which has obvious problems of temperature effects and so on. Sodium cyanide and sodium hydroxide have also been used successfully. It is important to have enough but not excessive alkalinity. About pH 10 is desired after combination with the acidic FCR and the samples. If the buffering capacity of the interconversion of carbonate and bicarbonate is not exceeded, evolution of free CO₂ bubbles to interfere with colorimetry is not ordinarily a problem.

The color may be developed more quickly at warmer temperature (Fig. 1), but, as discussed later, interferences may be greater. The blue color is relatively stable, and a standard, blank, and sample set read at 760 nm after 6 hr at room temperature gave slightly lower absorbance but similar analytical results to the 1-hr colorimetry, although with higher standard deviation. At higher temperatures, the loss of color with time is greater. Higher alkali levels also speed color development and its fading (Fig. 1).

The sample volume need not be 1.00 ml as long as the capacity of the linear range is not exceeded and conversion calculations and standards reflect the change. Microadaptation reduces costs. This procedure has been scaled down by a factor of five to a final volume of 20.0 ml.^{9,11} For that procedure, 2.00 ml of a 1:10 diluted sample (compared to the 100-ml procedure), 10.0 ml of FCR diluted 1:10, and 8.00 ml of 75 g/liter sodium carbonate gave the final 20.0 ml. Use of semiautomatic manual pipettes and syringe diluters and dispensers at the same final volume (20 ml) was also satisfactory with only a very slightly increased standard deviation. With adequate equipment, further size reduction is certainly possible.

Flow automation also is quite successful and an automated flow adaptation gave, with low-sugar samples, essentially identical values and a slightly lower coefficient of variation, but either heating to develop maximum color in a reasonable flow time or color measurement when it is still developing is required. Singleton and Slinkard¹¹ used an air-segmented flow system that delivered 0.42 ml of sample or standard per 20 ml final volume into 9.00 ml of dilution water, 5.29 ml of 1:5 dilution of FCR, and 5.29 ml of 100 g/liter sodium carbonate. A short 7-turn coil mixed the diluted sample, another of 28 turns followed FCR addition to provide adequate intermediate reaction time, and a third of 14 turns mixed in the sodium carbonate. It is considered important, as already discussed, to mix in the alkali well

¹¹ K. Slinkard and V. L. Singleton, *Am. J. Enol. Vitic.* **28**, 49 (1977).

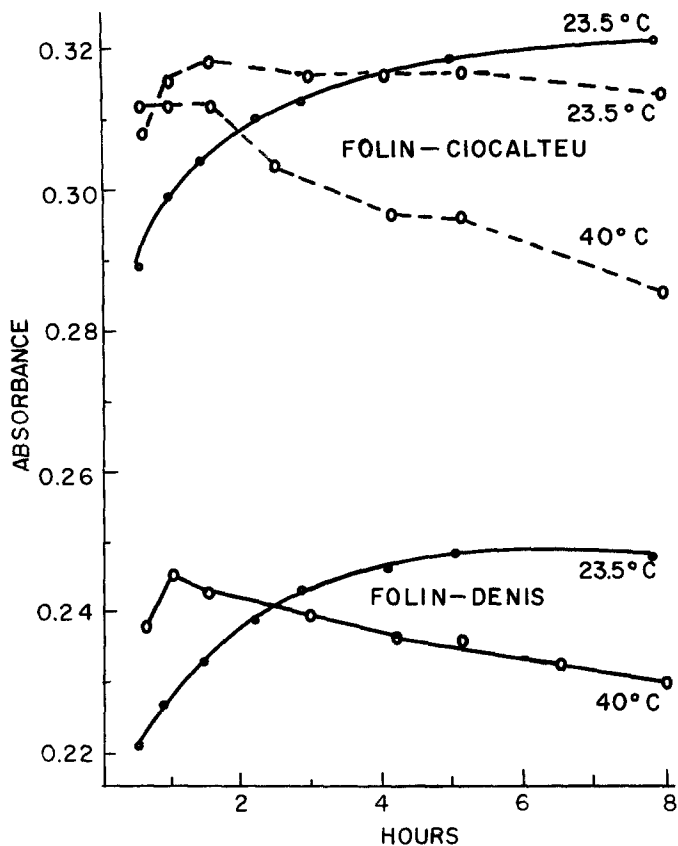


FIG. 1. Absorbance development from FCR and FDR with time at two temperatures and with 2.0 g/100 ml (solid lines) or 3.0 g/ml (dashed lines) sodium carbonate. Reproduced with permission from V. L. Singleton and J. A. Rossi, Jr., *Am. J. Enol. Vitic.* **16**, 144 (1965).

after the FCR to avoid premature alkaline destruction of the activity of FCR. The final mixture was passed through a 13-m coil in a 55° bath intended to provide about a 5-min delay and produce high color development similar to the manual method. Analyses were made at 40 samples or standards per hour. Absorbance was read at 760 nm in a flow cell with a 0.8-cm optical path.

Reading the absorbance manually while the color is still developing rather rapidly is impractical, but the reproducibility of flow automation makes it feasible. Celeste *et al.*¹² used this approach apparently at ambient

¹² M. Celeste, C. Tomas, A. Cladera, J. M. Estela, and V. Cerda, *Anal. Chim. Acta* **269**, 21 (1992).

temperature. Samples or standards (5–100 mg/liter of gallic acid) were injected, 60 samples per hour, into a flowing stream in 0.5-mm I.D. PTFE tubing. The sequence of addition was water, 0.7 ml/min; sample, 107 μ l each; FCR, 1.0 ml/min of a 1:10 dilution; and 1.0 ml/min of 0.5 *M* sodium hydroxide. Mixing coils were not indicated, but a reaction coil of 1 m was followed by reading the absorbance at 760 nm in a 1-cm, 18- μ l flow cell.

Standards and Blanks

A blank (phenol-free) solution fades rapidly from yellow to colorless unless the reagent has been partly reduced. The fact that the FCR is not stable under alkaline conditions emphasizes the importance of having sufficient excess present to react with all the phenols before it is destroyed.

Comparison standards as well as blanks are recommended to be included within each group of samples. Use of the absorbance produced under standard conditions as an "index" lacks the self-correction, easy transferability, and easy visualization built into the analysis if standards and blanks are used. Compounds used for standards have included tannic acid, gallic acid, catechin, tyrosine, and others. Most commonly, gallic acid has been used and the results are reported in milligram gallic acid equivalents (GAE) per liter. Earlier results on wines and spirits were considered "tannin" values because tannic acid was used as the standard. However, tannic acid from different preparations can vary, and other tannins cover a wide range of color yield per unit weight. Gallic acid is the significant phenolic unit in commercial tannic acid from oak galls. Gallic acid is equivalent on a weight basis if tannic acid is considered pentadigalloylglucose. The values in milligrams of tannic acid or gallic acid equivalents per liter on the same wine or spirit sample are very similar and relative values in a set of samples are directly comparable. Partly for this historical reason, gallic acid is widely used as the comparison standard. In addition, it is inexpensive, soluble in water, recrystallized easily from water, readily dried, and stable in the dry form. A stock solution is commonly made by dissolving 500 mg of gallic acid in a small amount of ethanol and diluting with distilled water to 1.00 liter. This will keep in a refrigerator for a day or two, but it is subject to slow oxidation and microbial attack. It is convenient to freeze portions suitable for the desired number of assays in oversized (so they do not break), screw-capped glass bottles. These may be held indefinitely and thawed as needed, taking care to mix any sublimed ice melt before opening.

The second most commonly used standard, (+)-catechin (mg CtE/liter), has advantages if flavonoid partitioning is being compared. As will be discussed under molar color yield, values can be interconverted.

The standard solution (500 mg GAE/liter) may be used as the top level in a series of standard dilutions, but its blue pigment production will be outside the absorbance range for satisfactory spectrophotometry in standard equipment. If standard and unknown samples prove to be too high, dilution of the blue color can indicate the proper phenol level of the samples, but they should also be diluted and reanalyzed. The measurable color yield should be linear or nearly so up to about 300 mg GAE/liter in the sample by the procedure described. Sample content is expressed most simply by direct computation from a plot prepared from standard dilutions with the same portion volumes as the samples rather than on the basis of the final reaction volume. The minimum detectable amount is of the order of 3 mg GAE/liter, especially if the sensitivity is extended by such techniques as cuvettes with longer light paths. Remember that oxidation, from air or otherwise, can alter stored samples, especially at low content.

Samples and Sample Preparation

Total phenol determination by FC (or FD) probably has been used most extensively with wines and spirits, but applications have been made with many kinds of samples, including fruit juices, plant tissues, sorghum grains, leather and antifeedant tannins, wood components, proteins, medicines, vanilla and other flavor extracts, olive oil, and water contaminated with phenols or treated with tannin to prevent boiler scale. Because of the potential for unusual problems or special interferences, some evaluation experiments should be conducted as new types of samples are analyzed.

Wines, brandies, whiskies, juices without appreciable insoluble pulp, and similar samples may be analyzed directly, with dilution if necessary, and consideration of potential interferences to be discussed shortly. Phenols from solid samples, of course, need to be converted to clear extracts suitable for colorimetry. An ethanol equivalent to 1 ml/100 ml of the final reaction mixture did not change the results in the normal assay with proper standards; however, the interference by free sulfur dioxide may be enhanced. Similarly, dilute aqueous solutions of other solvents unreactive in the assay may be usable (acetone, methanol, dimethylformamide have been reported), but testing and possibly preparing standards in the same solution are recommended.

An interesting application for intractable samples (e.g., solids present) is to conduct the reaction in suspension and measure the blue anionic pigment after its quantitative extraction into chloroform as a tetralkylammonium salt.¹³

¹³ A. Cladera-Fortaza, C. Tomas-Mas, J. M. Estrela-Ripoll, and G. Ramis-Ramos, *Microchem. J.* **52**, 28 (1995).

Chemistry of Reaction

The pertinent chemistry of tungstates and molybdates is very complex.¹⁴ The isopolyphosphotungstates are colorless in the fully oxidized 6⁺ valence state of the metal, and the analogous molybdenum compounds are yellow. They form mixed heteropolyphosphotungstates–molybdates. They exist in acid solution as hydrated octahedral complexes of the metal oxides coordinated around a central phosphate. Sequences of reversible one or two electron reductions lead to blue species such as (PMoW₁₁O₄₀)⁴⁻. In principle, addition of an electron to a formally nonbonding orbital reduces nominal MoO⁴⁺ units to “isostructural” MoO³⁺ blue species.

Tungstate forms are considered to be less easily reduced, but more susceptible to one-electron transfer. In the complete absence of molybdenum, phosphotungstates have been used to determine *ortho*-dihydric phenols selectively without including monophenols or *meta*-dihydric ones. Molybdates are considered to be reduced more easily to blue forms and electron migration is induced thermally. Mixed complexes as in the FC and FD reagents are intermediate, readily oxidizing monophenols and vicinal diphenols, but lacking in thermally enhanced electron delocalization. Detailed molecular and electronic structures of the blue reduction products are unclear and are likely to remain so in view of their complex nature. Long wavelength absorption maxima move from longer to short wavelengths and become more intense^{14,15} with greater reduction. Tungstate analogs had maxima at shorter wavelengths and lower molar absorptivities than molybdates, but followed similar trends. In solutions of increasing pH, a series of one-electron reductions can occur.

Blue products of phosphomolybdate reduction can have Mo⁶⁺ to Mo⁵⁺ ratios of 9.0 to 0.6. The 4 e⁻ reduced species is the most stable blue form and develops readily from mixtures of Mo⁵⁺ and Mo⁶⁺ in the necessary heteropolyphosphate forms. Absorption peaks are rather broad for the purer species of blue product, and the occurrence of several species can account for the very broad peaks found from FC and FD reduction (Fig. 2). Because of the breadth of these peaks and the fact that other components in biological samples do not absorb in this region, analysis can be carried out at a wide range of wavelengths, 760 nm generally being chosen for FC.

Although it is possible to form complexes between phenols such as catechol and phosphotungstates and molybdates, the phenol being oxidized by the FCR appears to have no other effect than to supply electrons. Different substrates do not appear to become part of the blue chromophore

¹⁴ M. T. Pope, *Prog. Inorg. Chem.* **39**, 181 (1991).

¹⁵ E. Papaconstantinou and M. T. Pope, *Inorg. Chem.* **9**, 667 (1970).

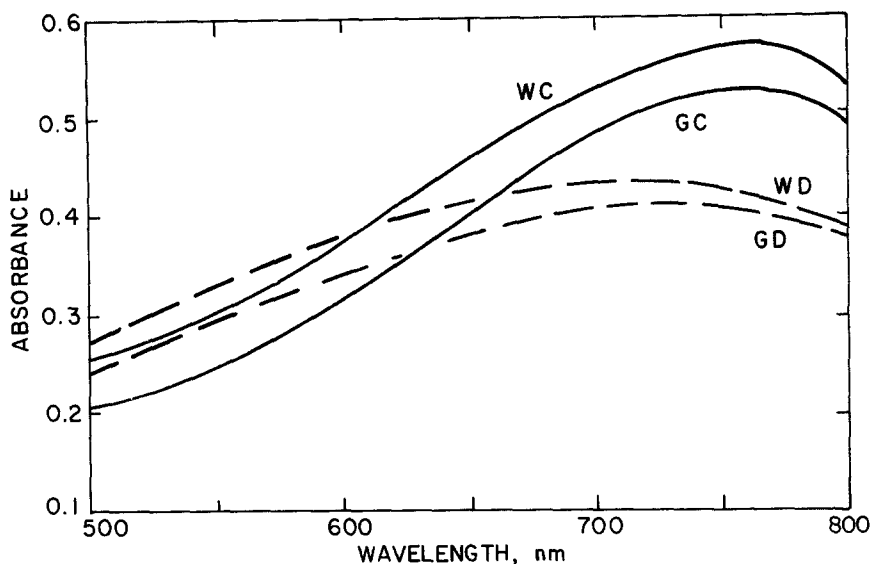


FIG. 2. Absorption spectra produced by wine (W) and gallic acid (G) with Folin-Ciocalteu (C) and Folin-Denis (D) reagents. Reproduced with permission from V. L. Singleton and J. A. Rossi, Jr., *Am. J. Enol. Vitic.* **16**, 144 (1965).

nor do the blue products appear different if generated by different substrates. This conclusion is based on the facts that the pertinent spectrum produced by different substrates is essentially the same (Fig. 2), gallic acid added to wine is recovered quantitatively,⁹ and the absorbance produced from a mixture of natural phenols of different classes is equivalent to the sum of their individual contributions.¹⁶

Molar Color Yields

The blue color generated at room temperature calculated as molar absorptivity related to the reacting substance has been determined¹⁶ for phenolic derivatives representing 29 monophenols, 22 catechols, 11 pyrogallols, 4 phloroglucinols, 9 resorcinols, 9 *para*-hydroquinols, 11 naphthols, 6 anthracenes, 17 flavonoid aglycones, 9 glycosides, 5 hydroxycoumarins, 7 aminophenols, and 19 nonphenolic substances. Selected data are shown in Table I.

¹⁶ V. L. Singleton, *Adv. Chem. Ser.* **134**, 184 (1974).

TABLE I
MOLAR ABSORPTIVITY IN FC ASSAY FROM SELECTED PHENOLS AND POTENTIALLY
INTERFERING SUBSTANCES^a

Compound	Molar absorbance (÷1000)	Free phenolic hydroxyls	Reacting groups	Molar absorptivity per reactive group
Phenol	12.7	1	1	12.7
<i>p</i> -Coumaric acid	15.6	1	1	15.6
Tyrosine	15.7	1	1	15.7
Catechol	22.5	2	2	11.2
Chlorogenic acid	28.9	2	2	14.4
DOPA	24.6	2	2	12.3
Ferulic acid	19.2	1	1+	19.2
Vanillin	14.9	1	1	14.9
Pyrogallol	24.8	3	2	12.4
Gallic acid	25.0	3	2	12.5
Sinapic acid	33.3	1	2	16.6
Phloroglucinol	13.3	3	1	13.3
<i>p</i> -Hydroquinone	12.8	2	1	12.8
Resorcinol	19.8	2	1+	19.8
(+)-Catechin	34.3	4	3	11.5
Kaempferol	29.6	3	2	14.8
Quercetin	48.3	4	3+	16.1
Quercitrin	44.8	4	3	14.9
Malvin	40.5	3	2+	20.2
4-Methylesculetin	31.1	2	2	15.6
<i>o</i> -Aminophenol	21.8	1	2	10.9
<i>p</i> -Aminophenol	12.5	1	1	12.5
<i>p</i> -Methylaniline	11.6	0	1	11.6
<i>o</i> -Diaminobenzene	21.4	0	2	10.7
Flavone	0.1	0	0	0.1
Flavanone	1.9	0	0+	1.9
3-Hydroxyflavone	3.5	0	0+	3.5
4-Hydroxycoumarin	0.1	0	0	0.1
Acetylsalicylic acid	0.2	0	0	0.2
D-Fructose	0.0	0	0	0
Ascorbic acid	17.5	0	1+	17.5
Ferrous sulfate	3.4	0	1-	3.4
Sodium sulfite	17.1	0	1+	17.1

^a Adapted with permission from V. L. Singleton, *Adv. Chem. Ser.* **134**, 184 (1974).

Other studies,^{17,18} allowing for changed conditions, including the use of FDR, have generally agreed very well in relative terms, even if not in absolute molar absorptivity, and have added new compounds. Under standardized conditions of the FC analysis, phenol itself gave 12,700 molar absorbance.¹⁶ Most other biologically likely monophenols tested gave similar or slightly higher (to 15,900) extinctions. Electron-attracting groups such as chloro, carbonyl, or nitro gave progressively less extinctions (4-chlorophenol, 12,100; salicylic acid, 8000; and picric acid, 0). Phloroglucinol and nearly all other *meta*-polyphenols reacted as monophenols, as did *para*-hydroquinols. Catechols and pyrogallol derivatives with free hydroxyls gave twice the color of monophenols in agreement with their *ortho*-quinone possibilities. Resorcinol behaved as a diphenol, but its derivatives generally tested as monophenols.

Flavonoids such as catechin closely approximated the sum of the color expected from the phloroglucinol A ring plus the reaction possibilities of their B ring (i.e., one plus two in this case). Flavonols such as quercetin, but not their 3-glycosides, gave more color than predicted, probably from participation of the enolic C ring. This idea is reinforced by the behavior of flavone and flavanones, which lack phenolic groups (Table I).

Generally, methoxyl substitution removed the reactivity of that phenolic group, but in some instances, particularly sinapic acid, there was indication of partial removal of a methyl group under assay conditions to generate additional free phenolic hydroxyls and additional blue pigment. Despite the alkaline conditions, ester and lactone formations involving phenolic hydroxyls appear to remain intact and inactive during the assay, at least at room temperature.

Reinforced with recovery from mixtures of phenols,¹⁶ these data show that a good first approximation of the contribution of a given weight of a specific compound to total color by the FC assay can be made by calculation taking into account the number of predicted reacting groups and the molecular weight compared to the gallic acid standard. Even better estimate can be made if the color yield of the specific compound in question is compared experimentally to gallic acid under the specific reaction conditions in use. It is unreasonable to expect that the total UV-HPLC peak area will correlate with total phenol by FC unless such calculations have been made. A high UV absorber may be a low contributor to the phenolic total and vice versa.

¹⁷ T. Swain and J. L. Goldstein, in "Methods in Polyphenol Chemistry" (J. B. Pridham, ed.), p. 131. Macmillan, New York, 1964.

¹⁸ M. Haug, B. Enssle, M. H. Goldbach, and K. Gierschner, *Ind. Obst. Gemueseverwert.* **69**, 567 (1984).

Interferences

The inclusiveness of the oxidation due to the FCR makes it unsurprising that the analytical result can include "interfering" substances in many crude, natural samples. If the possible interfering substances and their likely concentrations are known, efforts to limit the FCR assay to phenols can often be successful. To a degree the assay should be considered a measure of oxidizable substrates, not just phenols. In any case, results can be very useful, if interpreted properly.

Interferences can be of three types: inhibitory, additive, and enhancing or augmenting. Conceivably, inhibition could be from oxidants competing with the FCR, but such a reaction in samples should have been completed in advance. Air oxidation after the sample is made alkaline can certainly decrease phenols oxidizable by FCR. This is a reason why FCR addition ahead of alkali has been emphasized. If this is done, exposure is limited and is the same for samples and standards. Efforts to sparge and blanket with nitrogen have shown no significant effect. Inclusion of solvents other than water in the samples has sometimes inhibited color formation, but in practical usage the effect has been small or avoidable by solvent change or correctable by matching standards and blanks with the samples.

Additive effects are to be expected if unanticipated phenols or enols (e.g., additives or microbial metabolites) are present, as will also be the case with nonphenol FCR reactants. Aromatic amines as well as aminophenols are included in assays for total phenol (Table I). Tryptophan and other indoles react quantitatively with FCR, as do some purines. This has been known for a long time.^{5,8} Guanine (but not guanosine), xanthine, and uric acid reacted to give molar color yields with FCR equivalent to monophenols.^{19,20} Adenine and other purines and the common pyrimidines gave little color formation (about 1/50th as much). Alkaloids such as caffeine do not appear to have been tested. In wines and many other samples, insufficient of these compounds is present to contribute importantly in the assay for total phenols.

The reaction of proteins with FCR sometimes has been considered an interference, but this is somewhat unfair as most of the reaction is from tyrosine and tryptophan content. Further confusion has arisen from the fact that the Lowry method of protein determination^{21,22} uses FCR. However, in this method alkali is added and incubated with copper ions well before

¹⁹ T. E. Myers and V. L. Singleton, *Am. J. Enol. Vitic.* **30**, 98 (1979).

²⁰ M. Ikawa, C. A. Dollard, and T. D. Schaper, *J. Agric. Food Chem.* **36**, 309 (1988).

²¹ G. Legler, C. M. Müller-Platz, M. Mentges-Hettcamp, G. Pflieger, and E. Juelich, *Anal. Biochem.* **150**, 278 (1985).

²² C. M. Stoscheck, *Methods Enzymol.* **182**, 50 (1990).

the addition of FCR. This biuret-type reaction causes the conversion of nonphenolic dipeptides and larger into reactive enolic compounds and cuprous ions. The FC total phenol analysis procedure presented here does not add copper and avoids this reaction. Cuprous and ferrous ions can interfere, but not significantly, at the levels found in biological samples.

Cysteine gives a molar absorptivity with FCR of about 3500 (one-half or less of a monophenol), and hydrogen sulfide or other sulfhydryls such as glutathione are reported to contribute in total "phenol" assays with FCR, but this has not been studied fully. It is uncertain if this is only from direct oxidation of FCR or from hydroquinone regeneration to allow further oxidation by FCR. Reaction of a mercaptan with a quinone ring to produce a thioether substituted hydroquinone has been shown with all sulfhydryl compounds tested except dithiothreitol (DTT), which appears to give internal disulfide rather than quinone substitution.²³ Dithiothreitol by itself gives high FC color, but Larson *et al.*²⁴ reported that in the Lowry protein analysis addition of DTT after alkaline destruction of all of the FCR produces an enhanced color yield still proportional to the original protein content. Furthermore, color development is much more rapid to the final measurable level. This procedure deserves investigation for the non-Lowry FC assay without copper salts.

On a molar basis, sugars alone do not react appreciably at room temperature (Table I), but they may interfere with phenol analysis if the sugar level is high.^{11,25} At an elevated temperature, more interference is produced. This effect has been compensated for by applying standard corrections, separately determined corrections, or by preparing the standards in the same sugar concentration. Typical corrections¹¹ at room temperature and at 55° for various gallic acid levels are given in Table II.

The fructose effect was higher than that of glucose. Pectin at 5000 mg/liter added in equal volume to white wines gave only a 16-mg GAE/liter increment to the total in an automatic (heated) assay. Arabinose, galacturonic acid, and galactose had similarly low effects. The interference evidently is caused by the production in the strongly alkaline solution of enediol reductones from the sugar, a well-known reaction more intense with fructose. Under conditions of the assay and modest sugar levels, enediol production can be ignored at room temperature and is, even when heated, only a small part of the sugar present, but can be sizable compared to the content of phenols in sweet samples with low phenol levels. This may be why FC analysis of beers (made from cooked wort) has not been considered

²³ V. L. Singleton, M. Salgues, J. Zaya, and E. Trousdale, *Am. J. Enol. Vitic.* **36**, 50 (1985).

²⁴ E. Larson, B. Howlet, and A. Jagendorf, *Anal. Biochem.* **155**, 243 (1986).

²⁵ E. Donko and E. Phiniotis, *Szolez. Boraszat.* **1**, 357 (1975).

TABLE II
APPROXIMATE CORRECTIONS FOR INVERT SUGAR CONTENT FOR FC DETERMINATION
OF TOTAL PHENOLS^{a,b}

Apparent phenol content (mg GAE/liter)	A			B		
	Sugar content			Sugar content		
	25 g/liter	50g/liter	100 g/liter	25 g/liter	50 g/liter	100 g/liter
100	-5%	-10%	-20%	-20%	-30%	-60%
200	-5%	-8%	-20%	-20%	-25%	-38%
500	-4%	-6%	-10%	-17%	-24%	-38%
1000	-3%	-6%	-10%	-11%	-15%	-25%
2000	-3%	-6%	-10%	-10%	-13%	-17%

^a Adapted with permission from K. Slinkard and V. L. Singleton, *Am. J. Enol. Vitic.* **28**, 49 (1977).

^b FC conditions A = 25°, 120 min; B = 55°, 5 min. For instance, for a sample with 5.0% inverted sucrose under condition A, an apparent total phenol of 1000 mg GAE/liter should be corrected by 6% to give 940 mg GAE/liter.

as satisfactory as other methods. In the automated method,¹¹ 55° was chosen as the bath temperature because above that sugar began to participate over several days in browning reactions in white wine. Blouin *et al.*²⁶ nevertheless recommended 70° for 20 min after a careful statistical evaluation of FC assay conditions.

Ascorbic acid, an enediol, reacts readily with FCR and its presence must be considered. It reacts with polyphosphotungstate under acidic conditions (pH 3) in an assay that measures the blue color generated before the addition of alkali.²⁷ Verified with FCR, this procedure could be used to determine ascorbic acid before the phenols and its value then subtracted. In any case, appreciable blue formation from FCR before the addition of alkali indicates the presence of ascorbic acid or other very easily oxidized substance not requiring phenolate forms.

Ascorbic acid could have an augmentation effect on the amount of FCR reacting with the phenols present by reducing quinones as they form and prolonging the reaction. However, the FCR reaction with ascorbic acid appears sufficiently fast to prevent much of this effect. This may be one reason that a time lag is found desirable after combination of the sample and FCR and before the addition of alkali. Phenols oxidize little except as phenolates, and quinones should not form until after the ascorbic acid was already oxidized.

²⁶ J. Bloin, L. Llorca, F. R. Montreau, and J. H. Dufour, *Connaiss. Vigne Vin.* **4**, 403 (1972).

²⁷ F. W. Müller-Augustenberg and H. Kretzdorn, *Dtsch. Wein-Ztg.* **91**, 314 (1955).

The separate determination of ascorbic acid and correction of the total phenol accordingly are clearly important if enough is present to skew the results. In grape juice, little ascorbic acid remains after normal processing and less to none in wines or spirits unless it has been added.

The first oxidation product of ascorbic acid is dehydroascorbic acid. It evidently can accumulate as ascorbic acid reduces quinones from polyphenol oxidase action in grape juice/must production. Part of the interference in white wine analysis is from this source. Dehydroascorbic acid is not detected by usual ascorbic acid assays, but is enolic and can also react with FCR. Dehydroascorbic at 100 mg/liter has given FC values in heated flow automatic analysis equivalent to 45 mg GAE/liter. Separate analysis and subtraction of both forms are indicated, especially for white wines and other low phenol products.

Sulfites and sulfur dioxide react alone with FCR (Table I), but Somers and Ziemelis²⁸ showed that sulfite amplifies the reaction with phenols. This can be a serious problem in wines because not only is SO₂ often added for its antioxidant and antimicrobial effect, but yeast fermentation produces a small amount. Because the reaction appeared to be amenable to a correction formula, it was further investigated in our laboratories. The addition of sufficient acetaldehyde to fully bind the bisulfite present prevented this augmentation of the reactivity of phenols. Therefore it is believed the extra interference is caused by the rapid regeneration of oxidizable phenol by the sulfite, presumably by reduction or by substitution into the quinoid ring. Because duroquinone is not subject to augmentation by SO₂ and tetrabromocatechol and quinol are, the reduction of quinoids and not substitution is considered the likely mechanism.

Although sulfite in the aldehyde-bound form still reacts in the FC assay as it would alone, this reaction is reduced by about 50% to approximately a molar color production of 8000. In wines and most other modern food products, sulfites are low and are already in bound forms unless they have been freshly added, as was the case in the earlier work.²⁸

Using procedures such as selective distillation, freed sulfur dioxide also can be removed before FC assay.²⁹ D'Agostino³⁰ reported that SO₂ could be removed even from highly treated juices and concentrates to give FC values comparable to untreated material. Sugar corrections for a higher range were also given. Moutounet³¹ showed that interference by sugar and SO₂ is interactive and eliminated both by chromatographic treatment with

²⁸ T. C. Somers and G. Ziemelis, *J. Sci. Food Agric.* **31**, 600 (1979).

²⁹ G. Schlotten and M. Kacprowski, *Wein-Wiss.* **48**, 33 (1993).

³⁰ S. D'Agostino, *Vignevini* **13**(10), 17 (1986).

³¹ M. Moutounet, *Connaiss. Vigne Vin.* **15**, 287 (1981).

a dextran derivative (Sephadex LH-20). All phenols were retarded (by adsorption not gel exclusion) enough to allow washing through, by a small amount of water, of the polar interference compounds. The retarded phenols were eluted in 60% acetone. After removal of the acetone, FC analysis matched the value obtained on a model portion of the sample without sugar or sulfite.

The augmentation effect of sulfur dioxide was very different for different phenols. Phloroglucinol (Fig. 3), all three dimethyl phenols, and resorcinol gave an additive but no augmentation effect. Phenol, hydroquinone, gallic acid, and catechol derivatives gave a large augmentation over what would be expected from the individual components separately. The addition of acetaldehyde in more than equimolar amounts to the free SO_2 decreased this augmentation and sufficient prevented it (Fig. 4). Therefore a "swamping" concentration of acetaldehyde should be used.

Unless there were unusual amounts of free sulfite present (>250 mg/liter), the addition of 1000 mg/liter of acetaldehyde to the samples removed the augmentation effect and allowed simple subtractive correction for the residual oxidizability of the total bound bisulfite (Fig. 5). About 30 min at about pH 3 and room temperature is allowed for SO_2 binding by the

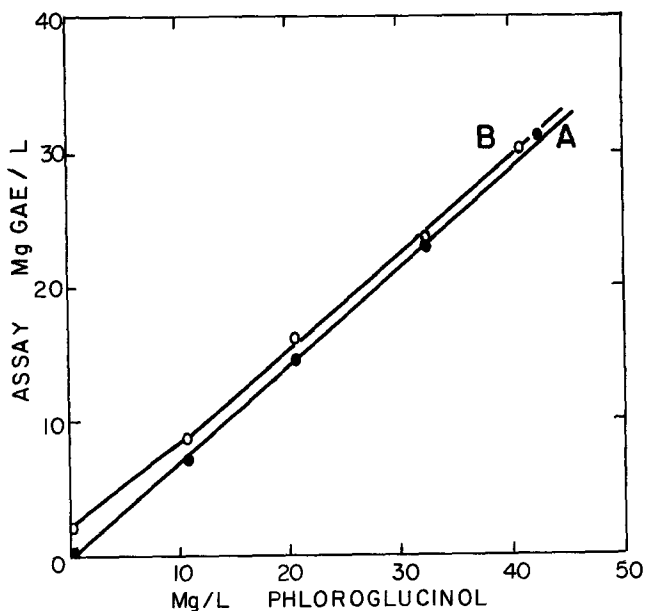


FIG. 3. Total phenol by Folin-Ciocalteu assay of phloroglucinol alone (A) and in the presence of 23 mg/liter of freshly added SO_2 (B).

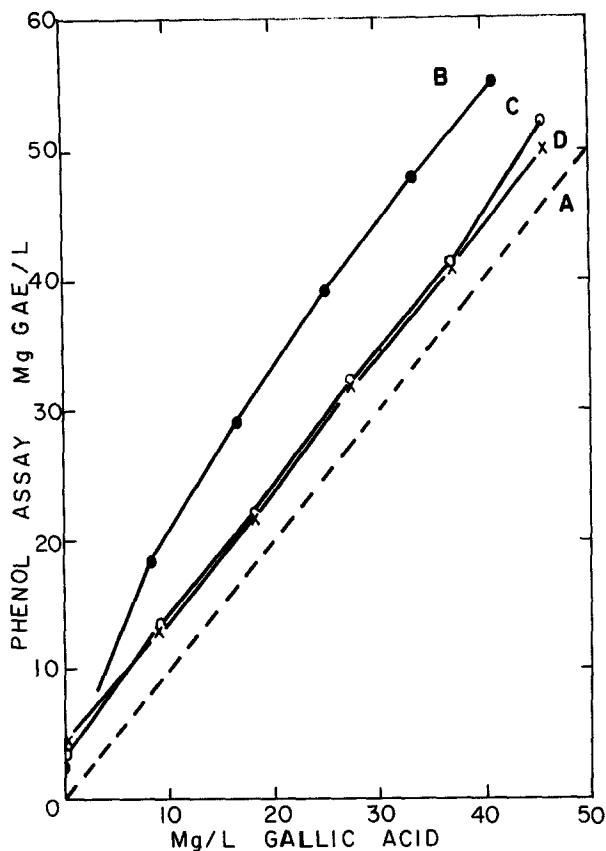


FIG. 4. Total phenol by Folin-Ciocalteu assay of gallic acid alone (A), with freshly added SO_2 at 24.2 mg/liter (B), 25.6 mg/liter SO_2 + 35 mg/liter acetaldehyde (C), and 25.3 mg/liter SO_2 + 79 mg/liter acetaldehyde (D).

acetaldehyde to occur. Under these conditions, precipitation caused by aldehyde bridging has not been a problem. The acetaldehyde itself added no FC color. Adjustments are made for dilution of course.

Automatic analysis (heated¹¹) without any corrections of a series of 36 dry white table wines made with differences in pomace contact from seven grape varieties gave total phenols by FC of 135–817 mg GAE/liter. Separate analysis of the same wines by a less inclusive UV spectral shift method gave values invariably 33–69% lower than FC values, but with a correlation coefficient of 0.975 between the two methods. There was some difference among varieties: Thompson Seedless giving greater and Palomino smaller differences between the two methods. These and other studies, including

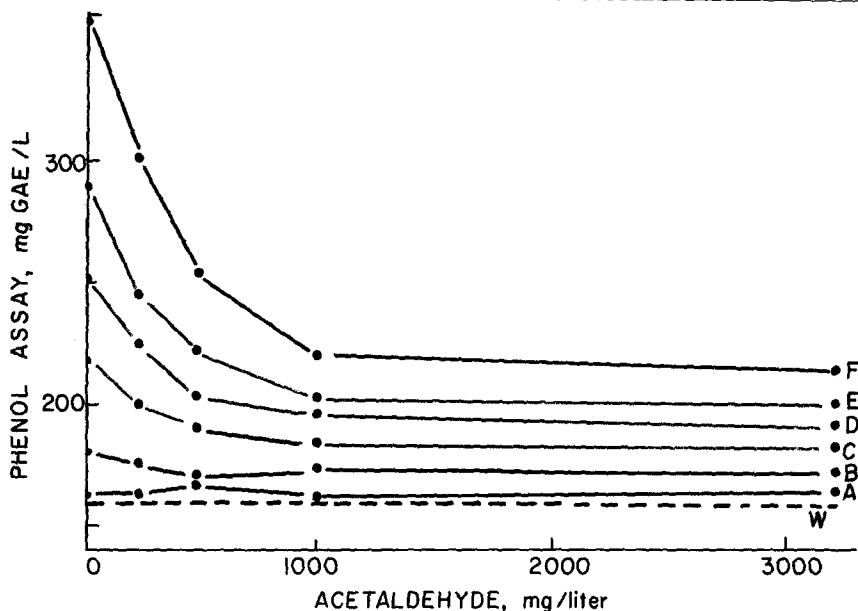


FIG. 5. Total phenol assay by Folin-Ciocalteu of a Riesling white wine alone (W) and with various levels of acetaldehyde following the addition of SO_2 in mg/liter of A = 25, B = 50, C = 100, D = 150, E = 200, and F = 300.

red wines at considerably higher total phenol levels, suggested total contributions of about 100 mg GAE/liter in typical light white wines without fermentable sugar. Such wines contain interfering substances in small but contributory amounts (bound sulfites, some purines, some dehydroascorbic acid, etc.) accounting for most, if not all, of the excess of the total phenol of FC over the sum of the readily identified individual phenols. Of course the comparison method may be at fault. Good relative values are possible routinely with FC and, with the consideration of potentially interfering substances and the levels present, more certainty is achieved.

Differential Methods and Variations

The total phenol by FC can be compared with values by other analyses on the same samples for subgroups such as tannins, flavonoids, anthocyanins, or phenolic acids. Among such other analyses are 520 nm absorbance for anthocyanins, leucoanthocyanidin conversion to cyanidin, and vanillin colorimetry for certain flavonoids. Especially for samples with high phenol content such as red wines, such comparisons have been useful, for example,

in studies of sample characteristics due to origin,³² storage changes,³³ and so on. Nevertheless, interpretation is not straightforward due to factors including overlap between substance categories and different assay characteristics. It is beyond the scope of this article to discuss these interrelationships. Several procedures make use of two FC analyses, one before and another after selective removal of some fraction. Only these will be discussed further.

Cliffe *et al.*³⁴ showed FC before and after laccase oxidation gave by difference the fraction oxidized by that enzyme (about 50% in apple juice). Jennings³⁵ compared samples analyzed by normal FC with those reacted 15 min in the presence of alkali before FCR addition. Molar extinctions were unchanged for compounds expected to act like monophenols except for quinol and considerably decreased for vicinal dihydroxyl phenols. The percentage decrease for specific phenols was up to 72%. It is unclear whether oxygen became limiting during the alkaline reaction. The method appears capable, with application over a wider series of controlled conditions, of clarifying the relative ease of oxidation among different phenols. Most of the other differential applications of two FC analyses have involved precipitation or adsorption reactions.

Flavonoids can be determined separately from other natural phenols by precipitation with formaldehyde. Phloroglucinol derivatives have electron-rich (nucleophilic) centers between the *meta*-hydroxyls. They react with formaldehyde and acetaldehyde in a strongly acid solution (pH <0.8) to produce first methylol substitution and then cross-linking leading to insoluble polymers. In natural samples the A rings of flavonoids are the only phloroglucinol derivatives likely to be present. This is the basis for a method to determine nonflavonoids separately that do not precipitate from flavonoids that do.³⁶

The procedure is to add 5.0 ml of 1:2 concentrated HCl and 5.0 ml of 8 mg/liter formaldehyde in water to 10.0 ml of a sample (about 2000 mgGAE/liter or less). Mix well and let stand 24 hr (or more) at room temperature under nitrogen. The nitrogen is probably unnecessary considering the high acidity, but is a simple precaution. Precipitation increases slightly as long as 72 hr, but the differences are small and small amounts of other phenols can be entrained. If the formaldehyde is too low, precipita-

³² E. Carruba, S. D'Agostino, B. Pastena, C. Alagna, and G. Torina, *Riv. Vitic. Enol.* **35**, 47 (1982).

³³ E. LaNotte and D. Antonacci, *Riv. Vitic. Enol.* **38**, 367 (1985).

³⁴ S. Cliffe, M. S. Fawer, G. Maier, K. Takata, and G. Ritter, *J. Agric. Food Chem.* **42**, 1824 (1994).

³⁵ A. C. Jennings, *Anal. Biochem.* **118**, 396 (1981).

³⁶ T. E. Kramling and V. L. Singleton, *Am. J. Enol. Vitic.* **20**, 86 (1969).

tion will be incomplete. If too high, the active sites (six and eight on catechin) can be excessively converted to methylol (soluble) derivatives and cross-linking decreased. Furthermore, the precipitate is soluble in 95% ethanol and appreciably in other aqueous solvents, including those from formaldehyde.

Formaldehyde at 40 mg per sample is about 15-fold maximum molar equivalent of the reactive phenols in higher samples in the indicated range and gave slightly more complete precipitation than did 120 mg per sample.³⁶ The formaldehyde solution was freshly prepared by dilution of 2.1 ml of 36% formalin to 100 ml. Note that the formaldehyde solution was listed erroneously as much too concentrated in some previous citations.^{37,38}

After 24 hr, the reaction mixture is centrifuged or the supernatant decanted, and, if not perfectly clear, filtered through a 0.45- μ m membrane filter. The residual nonflavonoid content is determined as usual by FC. Remember to compensate for the dilution compared to the original sample by either doubling the result or analyzing double the aliquot. Separately determine the total phenol by FC on the original, untreated sample. The total minus the nonflavonoid phenol (GAE or other matching units) equals the flavonoid content of the sample. The coefficient of variability of repeated nonflavonoid was about 3% and flavonoid values, because they involved two assays, were about double.

Obviously, if flavonoids are precipitated incompletely, the apparent nonflavonoids will be too high and if nonflavonoids are coprecipitated, too low. Less obviously, because sulfite augmentation would be swamped automatically by the formaldehyde, nonflavonoids by this procedure will be only subject to addition by sulfite, but unless the original total phenol was corrected to prevent any augmentation, the apparent flavonoid content could be too high. For a very low flavonoid content (<40 mg/liter), FC overestimated flavonoid content and had high variability, making it difficult to ascertain the completeness of removal of flavonoids by hyperoxidation of juices.³⁹ Also note that flavonoids with 4-carbonyl groups are deactivated and resistant to reaction with formaldehyde.

With some samples low in flavonoids, only a haze results from formaldehyde treatment. Conversely, anthocyanin glycosides, although flavonoids, are too soluble to precipitate completely unless sufficient condensed tannin or another less soluble phenol is available for cross-linking. Because phloroglucinol is precipitated completely by acidic formaldehyde, addition can

³⁷ V. L. Singleton, A. R. Sullivan, and C. Kramer, *Am. J. Enol. Vitic.* **22**, 161 (1971).

³⁸ V. L. Singleton, in "Methods of Plant Analysis" (H. F. Linskens and J. F. Jackson, eds.), p. 173. Springer Verlag, Berlin, 1988.

³⁹ V. Schneider, *Am. J. Enol. Vitic.* **46**, 274 (1995).

solve both problems. Faint pink may still remain in the strongly acidic solution even after phloroglucinol addition, but by spectral analysis this represents only a few milligrams of flavylum ions and an insignificant contribution to FC totals.

In general, phloroglucinol has not been added unless necessary for the two situations mentioned. When it is added, the procedure^{37,38} is modified as follows: 10.0 ml of sample, 2.0 ml of concentrated HCl, and 5.0 ml of 8 mg/ml formaldehyde for low flavonoid samples and 12 mg/ml for high ones are well mixed and allowed to stand at room temperature for 2–3 hr. Then 3.0 ml of 10 mg/ml phloroglucinol is mixed into high anthocyanin samples or 1.0 ml of the same plus 2.0 ml of water into those with low flavonoids. After a total reaction time of about 24 hr, the FC analysis is completed as before.

As is the case with most such reactions, the formation of a high amount of precipitate can adsorb phenols that were otherwise soluble. Because of the high acidity, it may be necessary to raise the concentration of the sodium carbonate used in the FC analysis, which should be checked under the specific conditions employed.

This flavonoid–nonflavonoid separation is not as precise as one would like. A few nonflavonoids precipitate slightly with formaldehyde,¹⁶ fisetin (resorcinol A ring) did not, and flavonoid glycosides alone are incompletely precipitated. However, the analysis has led to important findings. Nonflavonoid phenols in grapes and wines, mostly hydroxycinnamates, are relatively similar regardless of processing, whereas the flavonoid content rises greatly as pomace extraction occurs.³⁶ However, in apple juices a similarly wide range of total phenol content can be found, but the nonflavonoid content maintains a relatively constant fraction of the total.⁴⁰ Tannins extractable from oak barrels are hydrolyzable (not flavonoid) whereas those present from the grape are condensed (flavonoid) tannins. Because grape phenols other than flavonoid are low and relatively constant, oak barrel extraction during wine (or spirit) aging or tannic acid addition can be quantitated.³⁷ Scalbert *et al.*⁴¹ found good agreement with these results for a series of nonflavonoid tannins from various woods, but found that with added phloroglucinol these tannins and small, less soluble phenols were entrained appreciably.

Folin–Ciocalteu analysis has been applied before and after treatment with lead acetate, gelatin, hide powder, polyvinylpyrrolidone (PVP), polyamide, cinchonine, methylcellulose, sodium chloride, and ethyl acetate extraction. The first three can be passed over because lead acetate precipita-

⁴⁰ J. J. L. Cilliers, V. L. Singleton, and R. M. Lamuela-Raventós, *J. Food Sci.* **55**, 1458 (1990).

⁴¹ A. Scalbert, B. Monties, and E. Janin, *J. Agric. Food Chem.* **37**, 1324 (1989).

tion is known to be nonspecific and incomplete, gelatin is a soluble protein and could add FC color (albeit low compared to other proteins), and hide powder, the classical agent for tannin removal, is difficult to obtain, standardize, and apply. Nevertheless, the reported successful application of FC analysis after such diverse treatments shows its relative insensitivity to extraneous substances.

The objective in all these cases is to selectively remove a specific fraction of FC total phenols. The separation commonly sought is larger polyphenols from smaller polyphenols. The tannin group is heterogeneous in size (molecular weights between about 500 and 3000, larger is generally insoluble) and in chemical nature. They are of interest as a group because of their astringency, ability to precipitate proteins, inhibit enzymes, tan hides, and so on. These properties depend on a sizable molecule displaying a large number of free phenolic groups capable of producing numerous hydrogen bonds with proteins. This condition is met by both hydrolyzable and condensed tannins. Hydrolyzable tannins release gallic acid, ellagic acid, and related phenols when deesterified from their central sugar or other polar groups. Condensed tannins are flavonoid polymers capable of partial conversion to cyanidin (anthocyanogens, proanthocyanidins, leucoanthocyanidins). Reactions, particularly oxidation, during processing and storage modify these compounds to decrease or more commonly increase their size and incorporate anthocyanins or other units not otherwise present.^{6,42}

Saturation with sodium chloride is considered to precipitate from water the larger tannins (6–10 units). Those of lower polymeric size can be analyzed by FC in the salt solution and the lowest of all can be extracted into ethyl acetate. On a series of 27 red wines, plus 4 from heavily pressed pomace, and a red pigment extract, Margheri and co-workers^{43–46} found sodium chloride invariably precipitated less phenol than did treatment with cinchonine. The difference averaged 18% and the salt precipitation was also less consistent.

Single precipitation or adsorption procedures inevitably do not have sharp cutoff points among such diverse sizes and types of polyphenols. Tannin adsorbents polyamide or insoluble PVP do not readily adsorb nontannins such as gallic acid or (+)-catechin and also pass sugar and SO₂, allowing potential interference in FC analysis to remain. This would be expected to occur with other adsorbents and precipitants, causing such

⁴² H. P. S. Makkar and K. Becker, *J. Agric. Food Chem.* **44**, 1291 (1996).

⁴³ G. Margheri, D. Tonon, and F. Gottardi, *Vini Ital.* **18**, 337 (1976).

⁴⁴ G. Margheri and D. Tonon, *Riv. Vitic. Enol.* **30**, 376 (1977).

⁴⁵ G. Margheri, D. Tonon, and S. Inama, *Vini Ital.* **19**, 113 (1977).

⁴⁶ G. Margheri and G. Versini, *Vini Ital.* **21**, 83 (1979).

interference to be shown in unadsorbed or unprecipitated fractions. Particle size and method of preconditioning affect adsorption by these solids.⁴⁷

Mitjavila *et al.*⁴⁸ used soluble PVP (25,000–30,000 molecular weight), which alone would not precipitate tannins from beers, fruit juices, wines, and spirits, but did so in the presence of sufficient trichloroacetic acid. This precipitate centrifuged and washed with trichloroacetic acid is then taken up in water. Folin–Ciocalteu analysis, allowing for the acid, can then be applied to the dissolved precipitate as well as to the supernatant and the original sample. Gallic acid and (+)-catechin were found in the supernatant and not in the precipitate. Both grape seed (flavonoid) tannin and gallotannic acid were in the precipitate and not in the supernatant. About half of a leucocyanidin preparation was found in each. In a series of 2 beers (light and dark), 11 wines (red, pink, and white), and 4 spirits, the sum of FC values for the supernatant and precipitated fractions was 95.1–105.2% of the original FC total (average 97.8%). Probably due to effects of ascorbic acid, the assay of juices was less successful. There appears to be no obstacle to application of flavonoid precipitation to either fraction allowing four categories: hydrolyzable tannins, flavonoid tannins, nontannin flavonoids, and phenolic acids and other small phenols.

Peri and Pompei⁴⁹ adapted the precipitation of tannins by cinchonine sulfate at pH 7.0–8.0 followed by separating the supernatant. The precipitate is dissolved in aqueous ethanol containing 10% HCl.⁵⁰ The two resultant solutions are then treated with acid formaldehyde to precipitate flavonoids. Folin–Ciocalteu analysis of the redissolved cinchonine precipitate gives hydrolyzable tannins in the supernatant from formaldehyde precipitation and condensed tannins by difference. Similarly, analysis for total phenol by FC on the supernatant from cinchonine precipitation after formaldehyde precipitation gives in the new supernatant a measurement of the simple nonflavonoids and by difference nontannin flavonoids. Testing of knowns in each group at about 500 mg GAE/liter and mixtures of them gave very good agreement between the actual assays and the calculated concentrations. With a white wine having had no barrel age at 813 mg GAE/liter total FC phenol, the distribution was 42% condensed tannin, 0% hydrolyzable tannin, 33% nonflavonoid phenols, and 25% nontannin flavonoids.⁵¹

Among 27 red wines, averaged for the 9 lowest in phenol (864–1720 mg CtE/liter), the middle third, and the highest 9 (4800–5900 mg CtE/

⁴⁷ C. Pompei, C. Peri, G. Montedoro, E. Miniati, and N. Pasquini, *Ann. Technol. Agric.* **20**, 21 (1971).

⁴⁸ S. Mitjavila, M. Schiavon, and R. Derache, *Ann. Technol. Agric.* **20**, 335 (1971).

⁴⁹ C. Peri and C. Pompei, *Phytochemistry* **10**, 2187 (1971).

⁵⁰ A. Brugirard and J. Tavernier, *Ann. Technol. Agric.* **1**, 311 (1952).

⁵¹ C. Peri and C. Pompei, *Am. J. Enol. Vitic.* **22**, 55 (1971).

liter), the proportion of phenol precipitated by cinchonine increased and was, respectively, 32.1, 45.2, and 52.8% of the total FC phenol.⁴³⁻⁴⁶ Conversely, the proportion of the FC total phenol extractable by ethyl acetate decreased and was, respectively, 36.1, 24.7, and 14.6% of the original total. If one assumes no overlap between those phenols extracted by ethyl acetate (nonpolar, dimers and smaller) and those precipitated by cinchonine (tannins equivalent to trimers and larger), there was 30–33% of the total phenol attributable to nontannin, nonextractable phenols, i.e., polar compounds such as caftaric acid (caffeoyltartaric acid), and glycosides.

Montedoro and Fantozzi,⁵² with additions of highly methylated methylcellulose at double or more the tannin content, at pH 3–5, and with 100 g/liter of ammonium sulfate, did not precipitate hydroxy-substituted benzoates, cinnamates, simple phenols, flavonols, catechins, or phenolic glycosides, whereas cinchonine did precipitate portions of the last three. All three methods did precipitate gallotannic acid; methylcellulose more than PVP and slightly less than cinchonine. With model compounds⁵² and a wide series of beverages,⁵³ recovery of added tannic acid was good and repeated analyses had coefficients of variation of 1.3–13.1%. Formaldehyde precipitation on the supernatant after methylcellulose adsorption gives nontannin nonflavonoids and nontannin flavonoids by difference.

As just discussed, the three most widely applied methods to fractionate the large, FC-active phenols are precipitation with polyvinylpyrrolidone,⁴⁸ cinchonine sulfate,⁴⁹ or methylcellulose.⁵² Burkhardt⁵⁴ found on a must and red wines prepared from it with increasing degrees of pomace extraction that the three methods were equally suitable for laboratory analysis. For the three samples with the lowest phenol (530–780 mg CtE/liter), removal by cinchonine, PVP, and methylcellulose averaged, respectively, 1.3, 11.7, and 34.0% of the total FC phenol. For the three highest wines (2310–3280 mg CtE/liter), respective average removals were 48.7, 52.7, and 44.7%. From these data it appears that PVP had slightly greater affinity for the larger phenol molecules and methylcellulose a greater affinity in the dilute solution among mostly nontannin phenols. Unfortunately, formaldehyde precipitation was not included in these tests.

By a statistical study⁵⁵ of the analytical separation of total phenols by sodium chloride precipitation, formaldehyde precipitation, and comparison with other analyses and ratios, 93–100% of a group of red, pink, and wines made from a mixture of white and red grapes were classified correctly into

⁵² G. Montedoro and P. Fantozzi, *Lebensm.-Wiss. Technol.* **7**, 155 (1974).

⁵³ E. Polidori, G. Montedoro, and P. Fantozzi, *Sci. Tecnol. Alimenti* **4**, 157 (1974).

⁵⁴ R. Burkhardt, *Lebensmittelchem. Gerichtl. Chem.* **30**, 206 (1976).

⁵⁵ G. Santa-Maria, G. L. Garrido, and C. Diez, *Z. Lebensm.-Unters. Forsch.* **182**, 112 (1986).

one of these three types. With combinations of FC with cinchonine and formaldehyde methods, White Pinot wine⁵⁶ had 275 mg GAE/liter total phenol, apportioned as 39 mg GAE/liter of tannin (all condensed, presuming no addition of hydrolyzable), 175 mg GAE/liter as simple phenols (neither tannins nor flavonoids), and 61 mg GAE/liter of flavonoids not tannins.

Forty white wines just after first racking⁵⁷ gave range and average (in milligrams GAE per liter by FC): total phenols 202–1075, 672; grape tannin 30–465, 216; nontannin flavonoids, 10–507, 208; and phenolic acids 122–465, 216. The sum of the means of the three fractions is exactly the mean of total phenols, even though the range in values is great. Coefficients of variation among all the *samples* (not within assays) were 36.7% for total phenol, 31.2% for phenolic acids, 60.3% for tannins, and 68.2% for nontannin flavonoids, yet relationships held. The methods used were FC and successive precipitation with methylcellulose and formaldehyde.

Conclusions and Extensions

Analyses of the Folin–Ciocalteu type are convenient, simple, require only common equipment, and have produced a large body of comparable data. Under proper conditions, the assay is inclusive of monophenols and gives predictable (but variable by reactive groups per molecule) reactions with the types of phenols found in nature. Because different phenols react to different degrees, expression of the results as a single number such as milligrams per liter gallic acid equivalence is necessarily arbitrary. Because the reaction is independent, quantitative, and predictable, analysis of a mixture of phenols can be recalculated in terms of any other standard.

The assay in fact measures all compounds readily oxidizable under the reaction conditions and its very inclusiveness allows certain substances to also react that are either not phenols or seldom thought of as phenols (e.g., proteins). Judicious use, with consideration of potential interferences in particular samples and prior study if necessary, can lead to very informative results. Aggregate analysis of this type is an important supplement to and often more informative than reams of data difficult to summarize from techniques such as HPLC that separate a large number of individual compounds.

The predictable reaction of components in a mixture makes it possible to determine a single reactant by other means and to calculate its contribution to the total FC phenol content. Relative insensitivity of the FC analysis

⁵⁶ D. Villa, *Vignevini* **12**(4), 37 (1985).

⁵⁷ A. M. Gattuso, M. C. Indovina, and L. Pirrone, *Vignevini* **13**(4), 35 (1986).

to many adsorbents and precipitants makes differential assay before and after several different treatments informative. A balance sheet to estimate the fraction of the total assay due to additional, still unknown components is a use capable of expansion. Because the reaction is quantitative and all the ingredients other than the added phenol are inorganic, investigation of the exact nature of the FCR oxidation product(s) from specific phenols appears feasible. Such a study would give new insight into the nature of phenol oxidation in general as well as oxidation by FCR specifically.

[15] Size Separation of Condensed Tannins by Normal-Phase High-Performance Liquid Chromatography

By VERONIQUE CHEYNIER, JEAN-MARC SOUQUET, ERWAN LE ROUX,
SYLVAIN GUYOT, and JACQUES RIGAUD

Introduction

Condensed tannins, also called proanthocyanidins because they release anthocyanins when heated in acidic conditions, are ubiquitous plant components, consisting of chains of flavan-3-ol units (Fig. 1). Several classes can be distinguished on the basis of the hydroxylation pattern of the constitutive units. Among them, procyanidins, composed of (epi)catechin units (Fig. 1, $R_3 = H$), and prodelphinidins, deriving from (epi)gallocatechin (Fig. 1, $R_3 = OH$), are particularly widespread.

Within each class, monomeric units may be linked by C-4-C-6 and/or C-4-C-8 bonds (B type) or doubly linked, with an additional ether linkage (A type) and eventually substituted (e.g., glycosylated, galloylated). The structures of numerous oligomers have been elucidated.¹ However, the lower molecular weight proanthocyanidins are usually present in relatively low concentrations compared to polymers.² Besides, the degree of polymerization (DP) may vary greatly, as proanthocyanidins have been described up to 20,000 in molecular weight.³

Tannin properties, including radical scavenging effects and protein-binding ability, depend largely on their structure and particularly on the number of constitutive units (DP). Therefore, several methods have been

¹ L. J. Porter, in "The Flavonoids: Advances in Research since 1984" (J. B. Harborne, ed.), p. 23. Chapman and Hall, London, 1994.

² Z. Czochanska, L. Y. Foo, R. H. Newman, and J. L. Porter, *J. Chem. Soc. Perkin Trans 1* 2278 (1980).

³ E. Haslam and T. H. Lilley, *Crit. Rev. Food Sci. Nutr.* **27**, 1 (1988).